Natural Language Processing

Hard & Soft-Label for Hate Speech Classification

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Abstract

The automatic classification of hate speech is hindered by the inherent subjectivity of toxic language and the lexical overlap between hate speech and offensive slang. Standard supervised learning approaches frame this task using discrete "hard labels." However, analysis of hate speech datasets reveals a profound lack of consensus among human annotators. This project hypothesizes that training a model exclusively on deterministic labels forces overconfidence and discards crucial information regarding linguistic ambiguity. To address this, the project implements a soft-label training paradigm using Bidirectional Encoder Representations from Transformers (BERT). Instead of predicting a single class, the network is trained via a custom cross-entropy loss function to predict a probability distribution derived from human annotator votes. Experiments with standard baselines reveal that hard-label optimization artificially polarizes model confidence, leading to highly confident misclassifications on disputed text. In contrast, the soft-label BERT model successfully mirrors human uncertainty, minimizing Kullback-Leibler Divergence on ambiguous tweets. Following post-training threshold optimization, the soft-label approach performed better than the traditional hard-label fine-tuning on the minority Hate class (F1-score: 0.50 vs. 0.48) while maintaining high precision and recall on the majority classes, establishing that incorporating human doubt directly into the loss function improves both model calibration and discrete classification boundaries.

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1 Introduction

The volume of user-generated content on social media platforms renders manual moderation unfeasible. Automated hate speech classification is therefore necessary to enforce platform policies, identify targeted harassment, and moderate toxic content at scale. However, this classification task is complicated by the lexical overlap between genuine hate speech and language that is merely offensive.

Standard classification tasks utilize hard labels, assigning text to discrete categories such as Hate, Offensive, or Neither. However, exploratory analysis reveals a significant lack of consensus among human annotators, meaning forced discrete categorization discards critical information regarding linguistic ambiguity. This observation led to the project's core hypothesis: a model trained on rigid labels will fail to capture the nuances of hate speech. To address this, the project utilizes soft labels, representing the target variable as a probability distribution derived from annotator votes, thereby incorporating human uncertainty directly into the training process.

The report details a methodological progression. It begins with a TF-IDF and Logistic Regression baseline, followed by BERT feature extraction and Parameter-Efficient Fine-Tuning (LoRA). Finally, it compares full BERT fine-tuning under both hard and soft label paradigms. The models are evaluated using standard classification metrics (F1-score), latent space visualization (t-SNE), threshold-independent metrics (Kullback-Leibler Divergence), and explainability analysis (SHAP).

2 Dataset and Exploratory Data Analysis

2.1 Dataset Description and Annotation Rules

This project utilizes the Hate Speech and Offensive Language dataset introduced by Davidson et al. (2017) [1]. The dataset consists of tweets extracted using a lexicon of hate speech terms.

The ground truth labels were generated via crowdsourcing using the CrowdFlower platform. Annotators were provided with the definition of hate speech as "language that is used to express hatred toward a targeted group or is intended to be derogatory, to humiliate, or to insult the members of the group." They were instructed to categorize each tweet into one of three classes:

- **Hate Speech (Class 0):** Meets the strict definition of hate speech.
- **Offensive Language (Class 1):** Contains offensive terminology but does not target specific groups with hatred.
- **Neither (Class 2):** Contains neither offensive language nor hate speech.

Each tweet was reviewed by three or more annotators. The final hard label assigned to each tweet in the original dataset was determined by majority vote.

To ensure robust evaluation while preserving the class distribution, the dataset was stratified by class and divided into a 70% training set, a 15% validation set used for hyperparameter tuning and threshold optimization, and a 15% hold-out test set.

2.2 Data Cleaning

Social media text contains high levels of noise that interferes with tokenization and embedding alignment. A preprocessing pipeline was applied to standardize the input space. HTML entities were unescaped, and residual punctuation artifacts were stripped. To preserve contextual metadata without exposing the model to arbitrary strings, specific textual elements were replaced with standardized tokens. Mentions such as `@handle` were replaced with `user`, retweets (RT) with `retweet`, and hyperlinks with `url`.

For example, the raw input:

> "@ClicquotSuave: LMAOOOOOOOOOOO this nigga @Krillz_Nuh_Care http://t.co/AAnpSUjmYI" <bitch want likes for some depressing shit..foh

was normalized to:

> user LMAOOOOOOOOOOO this nigga user url bitch want likes for some depressing shit..foh

2.3 Class Imbalance and Annotator Consensus

Exploratory data analysis reveals a severe class imbalance. The dataset (comprising 24,783 total tweets) is heavily skewed toward Class 1 (Offensive) with 19190 instances, which is approximately 13 times larger than Class 0 (Hate, 1430 instances) and 4 times larger than Class 2 (Neither, 4163 instances).

More critically, analyzing the raw annotator vote counts exposes a fundamental flaw in utilizing majority-vote hard labels. By segmenting the dataset into unanimous cases, with 100% annotator agreement, and disputed cases, with mixed votes, a clear pattern emerges regarding human subjectivity.

As illustrated in **Figure 1**, the vast majority of instances ultimately labeled as Hate are disputed. This highlights a significant lack of human consensus

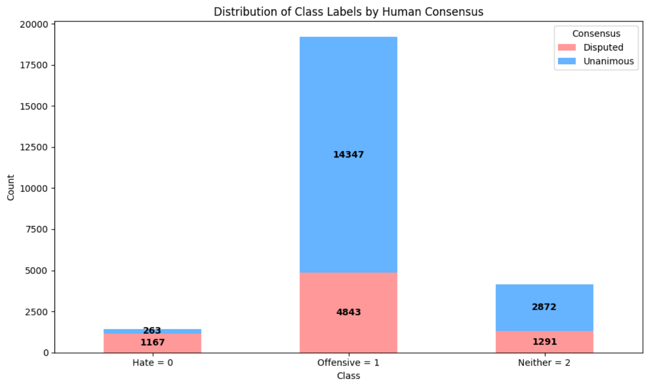

Figure 1: Class distribution highlighting the proportion of tweets where human annotators reached unanimous agreement versus those with disputed votes.

compared to the Neither and Offensive categories. The boundary between offensive slang and genuine hate speech is highly ambiguous, establishing the primary motivation for transitioning to soft-label training.

2.4 Linguistic Profiling

Analyzing the vocabulary across classes confirms the difficulty of the classification task. Visualizing the most frequent terms reveals a near-total vocabulary overlap between Class 0 and Class 1. Profanity and slurs are heavily utilized in both categories, meaning the mere presence of toxic words is insufficient for distinguishing hate speech from offensive language. Context and syntax are the primary differentiators.

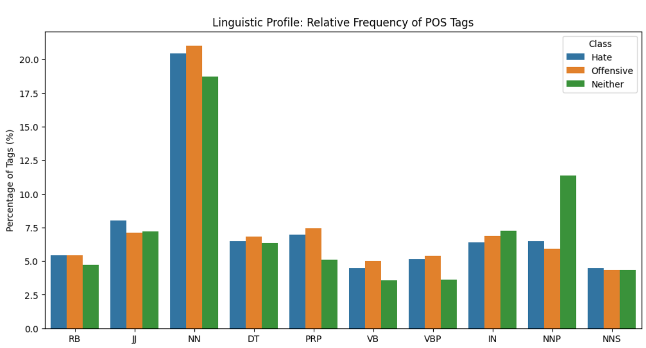

Figure 2: Relative frequency of Part-of-Speech (POS) tags across the three classes.

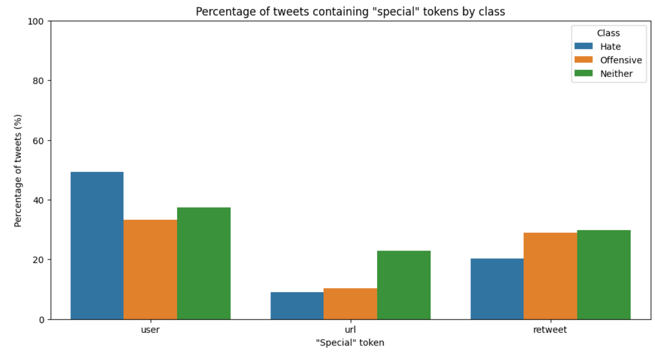

Figure 3: Distribution of contextual tokens (user, url, retweet) across classes.

Part-of-Speech (POS) tagging (**Figure 2**) indicates distinct syntactic profiles across the categories. Both the Hate and Offensive classes exhibit higher frequencies of personal pronouns (PRP) and adverbs (RB), suggesting a more informal, conversational, and often confrontational style. Conversely, the Neither class is characterized by a significantly higher usage of proper nouns (NNP), indicating discussions centered around specific entities, news, or neutral topics. Notably, while nouns (NN) are frequent across the toxic classes, the use of adjectives (JJ) is most pronounced in the Hate class, reflecting the descriptive and derogatory nature of hate speech.

Furthermore, analyzing the standardized metadata tokens (**Figure 3**) reveals that hate speech is highly targeted. Nearly 50% of tweets classified as Hate contain a user mention, indicating direct harassment. Conversely, the Neither class contains a significantly higher proportion of `url` tokens, characteristic of news sharing or general information dissemination rather than targeted abuse.

3 Baseline Models

Two baseline models were evaluated to establish a classification benchmark and justify the computational cost associated with deep learning: a traditional Bag-of-Words approach (TF-IDF) and a deep embedding extraction approach (frozen BERT). Because the dataset is severely imbalanced, standard accuracy is not a valid metric. Both baselines uti-

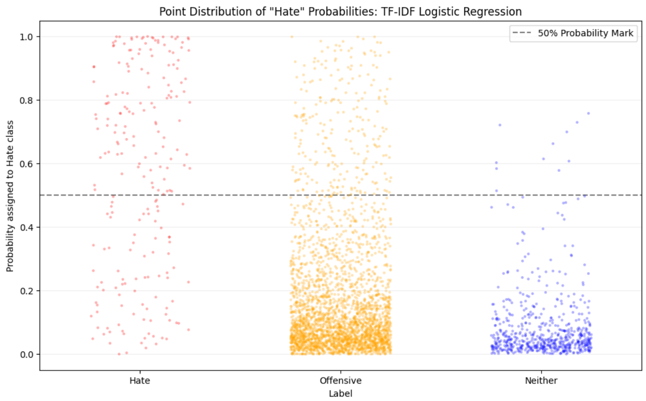

Figure 4: Probability distribution for the Hate class (TF-IDF).

lize Logistic Regression trained with balanced class weights, penalizing errors on the minority Hate class inversely proportional to its frequency. Model stability is validated using 10,000 bootstrap iterations to generate 95% confidence intervals (CI).

3.1 TF-IDF Baseline

Text was preprocessed using tokenization, stop-word removal, and lemmatization. A statistical phrase-detection model generated bigrams and trigrams, such as `white_trash`, with the vocabulary limited to 10,000 features. The model achieved a macro F1-score of 0.71, but the target Hate class achieved only 0.40 (95% CI: 0.35–0.44). Logistic Regression coefficients revealed a strong reliance on specific slurs.

As illustrated in **Figure 4**, the high dispersion of the Hate class indicates substantial uncertainty. The shared vocabulary between classes also generated numerous false positives, as the vectorizer ignored syntactic context.

3.2 BERT Feature Extraction

To address TF-IDF's contextual limitations, the frozen `bert-base-cased` model [2] was used to extract 768-dimensional [CLS] embeddings for Logistic Regression. This approach severely degraded performance, dropping the Hate F1-score to 0.28. The t-SNE projection in **Figure 5** displays a disorganized latent space with massive overlap, while the probability distribution plot reveals low overall confidence and a high rate of false positives. Because the BERT weights were frozen, the embeddings represent general linguistic structures learned during pre-training on Wikipedia and BookCorpus. The [CLS] token lacks the domain-specific tuning required to differentiate aggressive internet slang from genuine hate speech, confirming that task-specific fine-tuning is mandatory.

4 BERT Fine-Tuning and the Polarization Problem

To enable the model to learn domain-specific nuances, the pre-trained contextual embeddings must be adapted to the hate speech classification task. Two fine-tuning methodologies were implemented using hard labels: Parameter-Efficient Fine-Tuning (LoRA) and full fine-tuning. Both models utilized a custom Trainer class implementing a balanced, class-weighted cross-entropy loss function to mitigate the severe class imbalance.

4.1 Parameter-Efficient Fine-Tuning (LoRA)

Instead of updating all 109 million parameters of `bert-base-cased`, Low-Rank Adaptation (LoRA) [3] was utilized to optimize only the internal transformer projection matrices: the query, key, and value attention weights.

This approach significantly outperformed the frozen feature extraction baseline. The F1-score for the minority Hate class increased from 0.28 to 0.40 (95% CI: 0.35–0.44), matching the performance of the TF-IDF model but doing so with a better understanding of context, demonstrated by a reduction in false positives within the Neither category.

The t-SNE visualization (**Figure 5**, center panel) demonstrates an emerging structure in the latent space. Unlike the disorganized feature extraction baseline, the Neither class begins to migrate into distinct territory, and toxicity starts to form localized sub-clusters. However, a significant overlap between Hate and Offensive instances remains, indicating that updating only the attention weights is insufficient to fully disentangle the shared toxic vocabulary.

4.2 Full Fine-Tuning (Hard Labels)

Full fine-tuning updates all model parameters, allowing the network to completely realign its semantic representations. This approach yielded the highest traditional metrics. The macro F1-score reached 0.77, and the Hate class F1-score climbed to 0.48 (95% CI: 0.42–0.53). The Offensive and Neither classes achieved excellent F1-scores of 0.94 and 0.88, respectively.

As seen in the right panel of **Figure 5**, full fine-tuning successfully forces the classes into highly isolated clusters. However, an analysis of model confidence reveals a significant limitation inherent in applying the hard-label paradigm to subjective tasks.

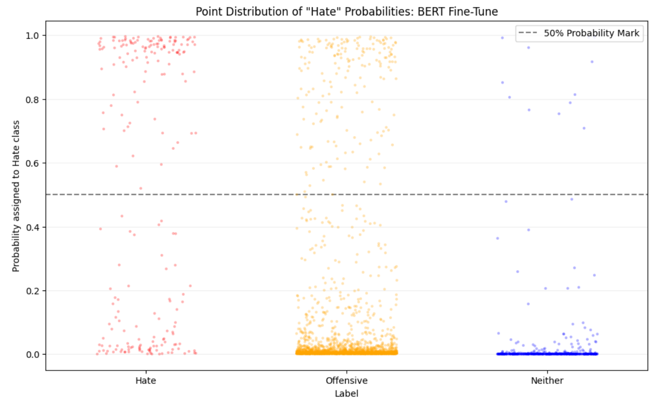

Figure 6: Probability distribution for the Hate class (Full Fine-Tune, Hard Labels).

As illustrated in **Figure 6**, the probability distribution is highly polarized (bimodal), with almost all predictions forced to the extremes 0.0 or 1.0. While this is a standard outcome of cross-entropy loss on hard labels, where the model is mathematically incentivized to maximize its confidence to reach zero loss, the effect is amplified by class weighting. By heavily penalizing errors on the minority class, the model is pushed to assign maximum probability to any instance containing toxic features.

Consequently, the model becomes confidently wrong. It successfully identifies many hateful tweets, but it also generates a significant vertical spray of false positives across the classification threshold. Because it has been trained to be deterministic, the model

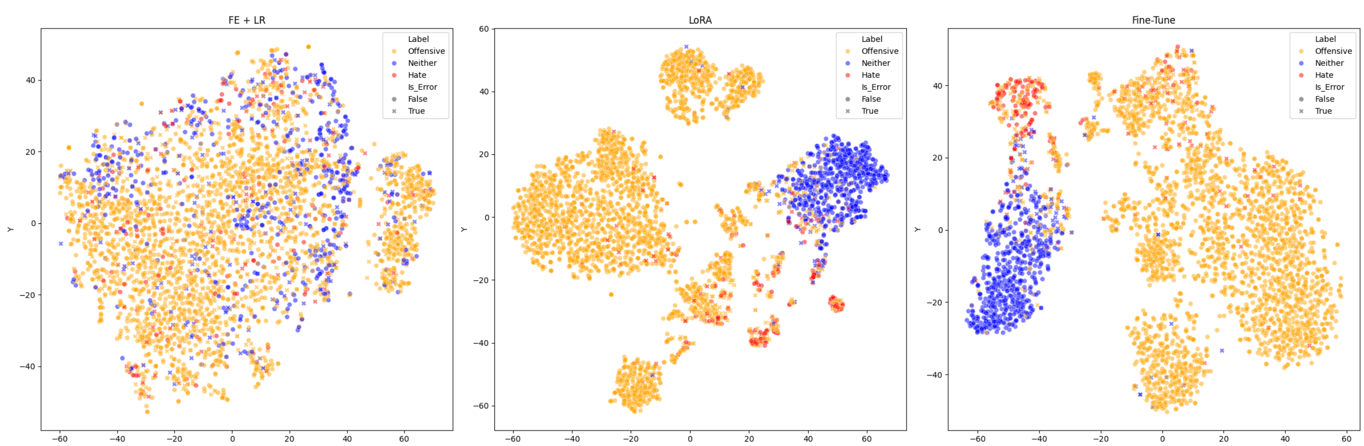

Figure 5: Evolution of the latent space (t-SNE) across the three hard-label BERT architectures: Feature Extraction (left), LoRA (center), and Full Fine-Tune (right).

frequently misclassifies colloquial, offensive language as hate speech with maximum confidence. For instance, it misclassifies "user user what do you mean nice try? I'm right nigga. You a faggot. If you wanna argue bout anything else continue" as Hate with near 1.0 certainty, despite 100% of human annotators agreeing it is merely offensive slang.

4.3 Human Agreement vs. Model Performance

Cross-referencing predictions with raw human consensus data revealed a substantial performance gap. Accuracy reached 96.39% when annotators unanimously agreed, but dropped to 73.53% in disputed cases, falling to 53.18% in the Disputed Hate subclass. The model struggled precisely where humans struggled. Forced to establish rigid decision boundaries in linguistically ambiguous contexts, the network lacked the capacity to express uncertainty. These findings suggest that modeling subjective language requires a probabilistic framework rather than a deterministic simplification.

5 The Soft Label Approach

To resolve the overconfidence introduced by discrete categorization, the training paradigm was shifted from hard to soft labels. Rather than forcing the model to predict a single, majority-vote class, the target variable was redefined as a probability distribution representing the proportion of human annotator votes.

For example, a tweet receiving one vote for Hate, two for Offensive, and zero for Neither is represented as the target vector $[0.33, 0.67, 0.00]$. A custom PyTorch Trainer was implemented utilizing a cross-entropy loss between the model's unnormalized logits and this target distribution. This function explicitly penalizes the model for deviating from the collective human judgment, effectively rewarding uncertainty when annotators were divided.

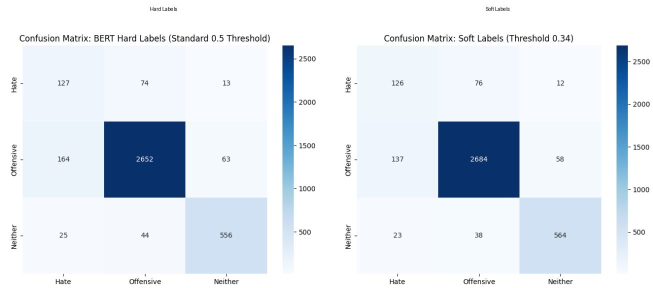

Figure 7: Confusion matrices evaluated on the test set comparing the Hard Label Full Fine-Tuning (left) and the Soft Label model with optimized threshold (right).

5.1 Threshold Optimization and Classification Performance

Applying class weights to a soft-label cross-entropy loss can distort the probability distribution, often leading to a systematic over-prediction of the minority class. To maintain the integrity of the learned human uncertainty, the model was trained on the original vote distributions without additional weighting. Instead, class imbalance was addressed post-training by optimizing the classification threshold on the validation set. This approach separates the model's ability to estimate probabilities from the final discrete classification decision.

An optimal decision boundary of 0.34 was identified for the Hate class. Applying this threshold to the unseen test set yielded a Hate F1-score of 0.50 (95% CI: 0.45–0.55). This outperforms both the TF-IDF baseline (0.40) and the hard-label full fine-tune model (0.48), while maintaining excellent performance on the Offensive (0.95) and Neither (0.90) classes.

The effect of this threshold optimization is summarized in **Figure 7**. Compared with the hard-label model, the soft-label approach reduces confusion around the minority Hate class while preserving the strong separation of the majority Offensive class.

As demonstrated in **Figure 8**, the probability distribution is no longer artificially polarized at 0.0 and 1.0. The model utilizes the entire probability spectrum, reflecting a calibrated understanding of linguistic ambiguity.

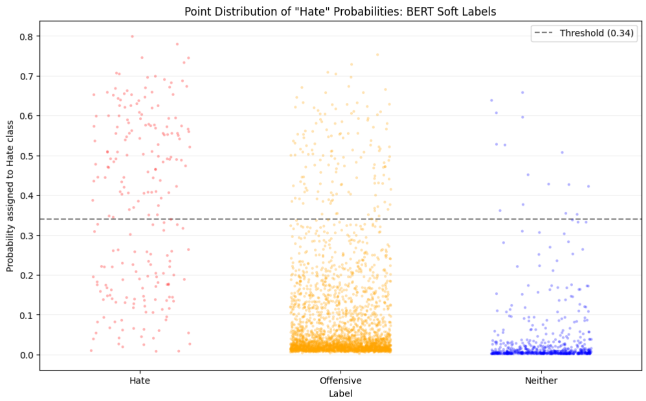

Figure 8: Probability distribution for the Hate class (Soft Labels). The distribution is no longer strictly bimodal, reflecting calibrated uncertainty.

5.2 Modeling Human Ambiguity: KL Divergence and MAE

While F1-scores evaluate discrete classification boundaries, they fail to measure whether the model accurately captures uncertainty. To evaluate the raw probability distributions, two threshold-independent metrics were utilized: Kullback-Leibler (KL) Divergence, measuring the informational distance between the human and model distributions, and Mean Absolute Error (MAE).

The soft-label model demonstrated superior alignment with human reasoning, particularly on ambiguous data. When comparing hard versus soft label performance on disputed subsets, the soft-label model consistently improved accuracy; for example, accuracy on Disputed-Hate increased from 53.18% to 56.07%.

More importantly, analyzing the lowest KL Divergence scores reveals instances where the model perfectly mirrored human doubt. For example, on the highly ambiguous tweet "user user fucking creepy ass nigguh lmao," the human annotators split their votes: Hate=1, Offensive=2, Neither=0, corresponding to $[0.33, 0.67, 0.00]$. The soft-label model predicted a near-identical distribution of $[0.34, 0.65, 0.00]$, yielding a KL divergence of just 0.0026 and an average error per class of 0.9%. This confirms the hypothesis: the network is capable of learning the exact boundaries of human disagreement when provided with a probabilistic target.

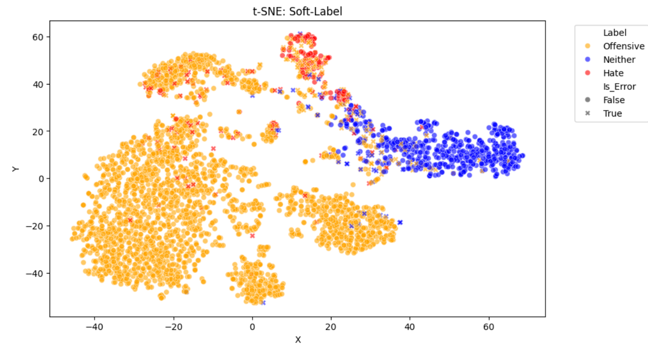

Figure 9: t-SNE projection of the [CLS] embeddings after Soft-Label fine-tuning.

5.3 Error Analysis of Soft Labels

Analyzing the highest KL Divergence scores isolates the model's remaining failure modes. The model severely misunderstands human intent (KL > 2.0) primarily when highly localized slang or non-standard syntax alters the semantics of a known slur.

For example, on the tweet "I didnt buy any alcohol this weekend, and only bought 20 fags. Proud that I still have 40 quid tbh," human annotators recognized the UK slang for cigarettes and voted Neither $[0.00, 0.33, 0.67]$. The model, however, heavily weighted the slur and output a toxic distribution $[0.42, 0.57, 0.01]$. These errors indicate that while soft labels successfully calibrate confidence regarding general internet toxicity, the model still requires broader contextual pre-training to resolve specific demographic vernaculars.

Finally, the t-SNE visualization of the soft-label embeddings (**Figure 9**) demonstrates a balanced latent space. It retains the localized clusters for toxicity seen in the hard-label fine-tuning but avoids the extreme, artificial isolation of data points, accurately reflecting the continuous spectrum of offensive language.

6 Casing Impact: Cased vs. Uncased Architectures

Social media frequently utilizes capitalization to convey aggression. To determine if this typographical metadata improves classification, all experiments were replicated using `bert-base-uncased`. Comparative analysis revealed that performance differences were marginal and generally fell within the 95% bootstrap confidence intervals. For full fine-tuning with soft labels, the uncased model achieved a slightly higher Hate F1-score (0.51) compared to the cased model (0.50). Regarding distributional alignment (KL Divergence and MAE), the models performed almost identically on ambiguous text. A theoretical trade-off is hypothesized: while cased models can capture capitalized aggression, uncased models benefit from reduced vocabulary sparsity by collapsing case variations of slurs into single tokens, which may be advantageous for smaller datasets. Ultimately, the choice between architectures proved secondary to the paradigm shift from hard to soft labels.

7 Explainability and Error Analysis

To understand the internal decision-making processes of the fine-tuned architectures, SHapley Additive exPlanations (SHAP) [4] were utilized. By analyzing the marginal contribution of individual tokens to the model's final output, we can observe how the soft-label training paradigm alters feature importance compared to the hard-label model.

7.1 SHAP Analysis: Modeling Human Doubt

When analyzing the best predictions, defined as instances with the lowest KL Divergence on disputed tweets, SHAP visualizations reveal how the soft-label model successfully distributes probability mass. In colloquial text where toxic slurs are used conversationally, such as "user user fucking creepy ass nigguh lmao," human annotators split their votes between Hate (1) and Offensive (2). The soft-label model mirrored this almost perfectly: $[0.34, 0.65, 0.00]$. SHAP analysis demonstrates that while the roots of the slurs, including *creepy*, *ni*, *ass*, and *fucking*, push the prediction toward the Hate class, their magnitudes are constrained. Rather than allowing a single toxic token to monopolize the decision and force a 1.0 confidence score, as seen in the polarized hard-label model, the soft model evaluates the entire syntax, allowing the surrounding colloquialisms to pull a proportionate amount of probability mass into the Offensive class.

7.2 Remaining Failure Modes

Despite superior calibration, instances with the highest KL Divergence exposed persistent limitations. The first failure mode involved demographic vernacular, where toxic strings carried innocuous regional meanings, such as "bought 20 fags" as UK slang for cigarettes. While human annotators correctly recognized the context, SHAP analysis revealed that the subwords *f* and *ag* exerted a strong positive influence toward the Hate class, overpowering contextual cues. The second failure mode involved subword tokenization collisions. Unfamiliar slang such as "jiggers" was fragmented by WordPiece into *ji* and *gger*. The model heavily weighted the *gger* subword because of its association with a severe racial slur, triggering a false positive. Resolving these errors likely requires broader contextual pre-training or customized tokenization strategies.

8 Conclusion

This project addressed the subjectivity of hate speech classification by transitioning from deterministic hard labels to probabilistic soft labels. Baseline models and standard full fine-tuning showed that hard-label optimization on imbalanced, subjective data induces severe overconfidence, producing highly confident false positives on disputed text. By implementing a custom cross-entropy loss over the distribution of human annotator votes, the soft-label BERT model mitigated this overconfidence. Post-training threshold optimization at 0.34 yielded the highest F1-score for the minority Hate class (0.50) without sacrificing majority-class performance. Furthermore, KL Divergence and SHAP-based explanations indicate that the soft-label network successfully learned the boundaries of human disagreement. The results support the core hypothesis: modeling subjective human language requires a probabilistic representation of uncertainty rather than rigid categorical labels.

Limitations and Future Work: Despite the improved calibration, explainability analysis revealed that the model remains vulnerable to subword tokenization collisions (e.g., WordPiece fragmenting unfamiliar slang) and lacks the cultural context to decode harmless regional vernaculars (e.g., UK slang). Future work should address these limitations by exploring custom, domain-specific tokenization strategies and conducting further pre-training on corpora rich in demographic and regional slang to better disentangle severe toxicity from colloquial internet jargon.

A Class-Specific Word Clouds

The following word clouds summarize the most frequent vocabulary observed within each class after preprocessing. They provide a qualitative view of the lexical overlap between Hate Speech and Offensive Language, as well as the more neutral vocabulary characterizing the Neither class.

Figure 10: Word cloud for Class 0: Hate Speech.

Figure 11: Word cloud for Class 1: Offensive Language.

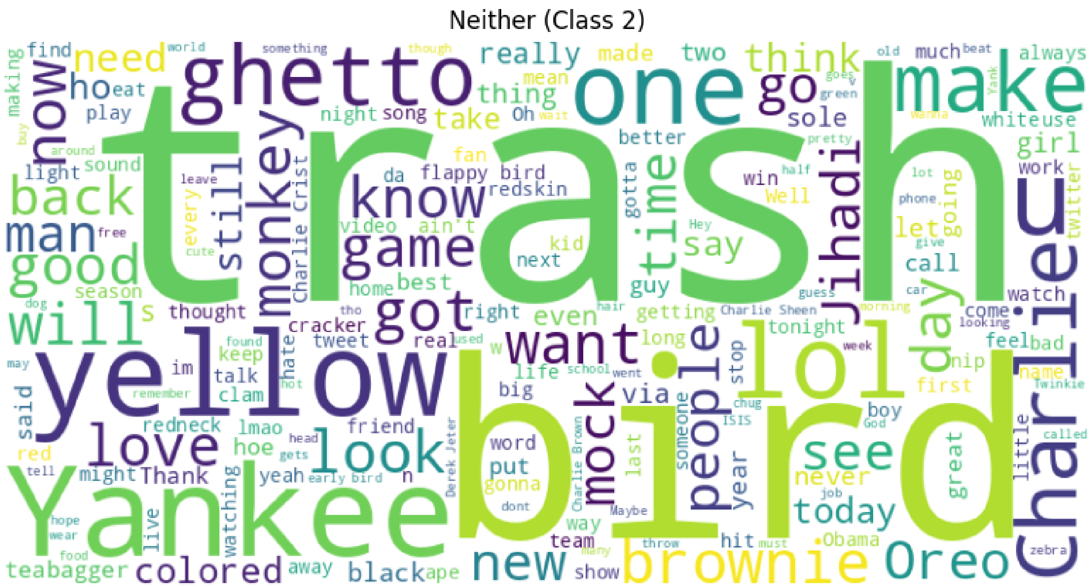

Figure 12: Word cloud for Class 2: Neither.

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